SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	01/09/2026	**Duration:**	60 minutes

Code of Conduct

I ___M. SANA FATHIMA(192371082)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _M. SANA FATHIMA

Annexure A – Research Paper Review / Literature Survey

ASSIGNED PROBLEM – E COMMERCE ORDER FULFILLMENT PLATFORM

Introduction

E-commerce order fulfillment is the end-to-end process through which an online order moves from customer placement to inventory confirmation, warehouse allocation, picking, packing, dispatch, last-mile delivery, delivery confirmation and, where required, return processing. The literature shows that fulfillment is not a single warehouse task; it is a networked decision problem involving inventory, order allocation, picking, batching, transportation, delivery promises and exceptions. The proposed E-Commerce Order Fulfillment Platform therefore treats fulfillment as an integrated software system rather than a collection of isolated activities.

Problem Statement

E-commerce platforms must fulfill a growing number of small, time-sensitive and geographically distributed orders while maintaining product availability and controlling fulfillment cost. A practical order fulfillment platform must decide where an order should be fulfilled, how items should be picked and consolidated, when orders should be released, which carrier or delivery route should be selected, and how exceptions should be handled. These decisions are interdependent: improving warehouse throughput without considering delivery capacity can create downstream congestion, while optimizing delivery alone may increase picking or inventory costs.

Objectives

- Review credible research on e-commerce fulfillment, warehouse operations, order allocation, batching and delivery.
- Compare the methodologies, tools, datasets and findings of selected research papers.
- Identify limitations in approaches that optimize only one stage of the fulfillment chain.
- Identify a research gap around integrated, real-time, exception-aware fulfillment.
- Define future directions using analytics, optimization, AI-assisted decisions and event-driven software architecture.
- Translate the literature findings into expected platform deliverables and measurable outcomes.

Literature Survey – Selected Research Papers

Paper / Year	**Research Problem**	**Methodology / Tools**	**Data / Setting**	**Key Finding**	**Limitation**	**Research Gap**
Boysen, de Koster & Weidinger (2019)	E-commerce warehousing	Survey of warehousing systems, automation, batching, zoning and sorting	Literature review; warehouse systems	E-commerce creates small, time-critical orders and volatile workloads; automation dynamic	Survey-level study; not an end-to-end platform.	Need stronger integration of warehouse decisions with downstream delivery.

Bansal et al. (2021)	Stock-to-picker order fulfillment	Closed queuing-network models; static vs dynamic batching; simulation validation	Waveless fulfillment model; simulation	Dynamic batching improves throughput; multi-line orders showed 37–43% throughput improvement in reported experiments.	Focuses on batching/pick station rather than complete fulfillment lifecycle.	Need real-time coordination between batching, inventory and delivery.
Kandula, Krishnamoorthy & Roy (2021)	Efficient e-commerce order delivery	Predictive modeling + VRPTW decision-support framework	Two heterogeneous real-world e-commerce datasets	Data-driven delivery scheduling can reduce missed deliveries and delivery costs.	Strong last-mile focus; customer availability information is difficult to maintain.	Need integration with upstream allocation and warehouse state.
Urquhart et al. (2022)	Online grocery fulfillment and last-mile capacity	Analysis of capacity constraints in store-based fulfillment	Great Britain grocery fulfillment context	Store-based picking and delivery capacity create operational constraints as online demand.	Sector and geography specific.	Need dynamic capacity-aware allocation across multiple fulfillment nodes.

Individual Paper Reviews

Warehousing in the E-Commerce Era – Boysen, de Koster & Weidinger (2019)

This survey examines warehousing systems used by online retailers. It highlights the operational characteristics of e-commerce: large assortments, many small orders, time-critical picking and highly variable workloads. The authors discuss automated picking workstations, robots, AGV-assisted systems, dynamic order processing, batching, zoning and sorting. The work is important for the proposed platform because it establishes that warehouse software must support both automation and flexible order processing. The paper also makes clear that traditional picker-to-parts approaches can become inefficient when orders are small and delivery promises are tight.

Performance Analysis of Batching Decisions – Bansal et al. (2021)

Bansal and colleagues analyze stock-to-picker fulfillment using closed queuing-network models and simulation. Their results indicate that dynamic batching can outperform static batching in throughput, with reported improvements of 37–43% for multi-line orders under the studied settings. This supports a platform design in which batching is treated as a dynamic decision rather than a fixed operational rule.

The main limitation is scope: the research concentrates on the pick-station environment. In a full platform, a batch that looks efficient for picking may be poor for delivery if it creates a late outbound wave or an oversized carrier load. Therefore, the research motivates a cross-module design in which batching decisions consume delivery and service-promise information.

A Prescriptive Analytics Framework for Efficient E-Commerce Order Delivery – Kandula, Krishnamoorthy & Roy (2021)

This work focuses on last-mile delivery and proposes a decision-support framework that combines predictive modeling with vehicle-routing-with-time-windows concepts. The motivation is that optimizing only route distance can cause missed deliveries when customer availability is not considered. The study uses two heterogeneous real-world e-commerce datasets and reports improvements in delivery success and delivery cost.

Its major contribution to the proposed platform is the idea of moving from descriptive tracking to prescriptive decisions. A fulfillment dashboard should not only show that an order is late; it should identify which orders, routes or delivery slots require intervention. The limitation is that the delivery model is downstream of warehouse and inventory decisions. The proposed platform closes this gap by passing warehouse readiness, inventory allocation and promised delivery windows into delivery planning.

Enhancing E-Grocery Order Fulfillment – Ekren et al. (2024/2025)

Ekren and colleagues study fulfillment-center allocation using customer and e-grocery data. The work evaluates product availability, cost and emissions and finds that policies informed by historical purchases, product popularity and delivery patterns can outperform policies that do not use those data. The study also identifies an important trade-off: allocating based on proximity can reduce delivery distance but may increase the risk that a selected product is unavailable.

This paper supports multi-objective allocation in the proposed platform. Instead of selecting the nearest warehouse only, the allocation engine should consider inventory availability, distance, service promise and cost. The remaining research opportunity is to operationalize these decisions in a continuously updated software system with exception handling and customer-facing visibility.

Synchronizing Order Picking and Delivery – Ouyang et al. (2024)

Ouyang and colleagues model the interaction between order picking and outbound delivery using a Markov Decision Process and a CNN-based prediction component. Their study reports that synchronization can improve performance under limited picking resources and demonstrates the value of predicting near-future delivery decisions.

Integrated and Recent Fulfillment Research – 2025–2026

Recent research increasingly treats fulfillment as a network decision problem rather than a single warehouse activity. Work on multi-item online fulfillment considers real-time warehouse assignment under inventory constraints, while recent middle-mile research examines stochastic optimization to reduce cost and pressure on the last mile. Newer quick-commerce research also combines partner selection and order optimization under uncertain demand. These studies collectively indicate a shift toward real-time, multi-objective and data-driven fulfillment.

Comparative Analysis

The comparison shows five major technology directions: warehouse automation and dynamic batching; predictive and prescriptive delivery analytics; multi-node inventory-aware allocation; integrated picking/delivery synchronization; and multi-objective optimization under uncertainty. Analytical models can produce strong decision quality, but production systems require event processing, APIs, databases, observability, exception management and user interfaces.

Parameter	Existing Literature	Proposed Platform
Fulfillment scope	Often optimized by stage	End-to-end order lifecycle
Decision style	Optimization, simulation, prediction, MDP	Hybrid rules + optimization + analytics
Inventory	Inventory constraints studied in allocation papers	Real-time stock, reservations and allocation visibility
Warehouse	Batching, picking and automation	Dynamic queue, pick/pack status and capacity
Delivery	Routing, time windows and partner decisions	Carrier assignment, ETA, exceptions and tracking
Data	Historical datasets, simulations, case data	Operational events + historical analytics
Scalability	Model-specific assumptions	Modular services and multi-node design
Exceptions	Often outside model scope	First-class workflow with escalation
Customer visibility	Limited in optimization studies	Order status, ETA and delivery updates
Sustainability	Considered in selected allocation studies	Optional KPI: distance/emissions per order

Critical Analysis and Research Gap

The literature is strong in individual decision areas but remains fragmented from a software-system perspective. Warehouse research explains how picking, batching and automation affect throughput; delivery research explains routing, customer availability and time windows; allocation research explains inventory-aware fulfillment-center selection. However, these decisions are frequently modeled separately or under narrow operational assumptions. This separation creates a practical gap because real fulfillment systems must make decisions while the state of inventory, warehouse capacity, carrier availability and customer delivery promises changes continuously.

A fourth gap is the combination of sustainability and service quality. Some recent research explicitly considers cost and emissions, but a practical platform can expose these as operational KPIs alongside on-time delivery, fill rate, picking productivity and return rate.

Proposed Platform Workflow

Figure 1. Proposed end-to-end fulfillment workflow. The screenshot/figure is a prototype representation created for this literature-survey deliverable; it is not claimed to be a screenshot from an existing commercial platform.

Proposed Architecture

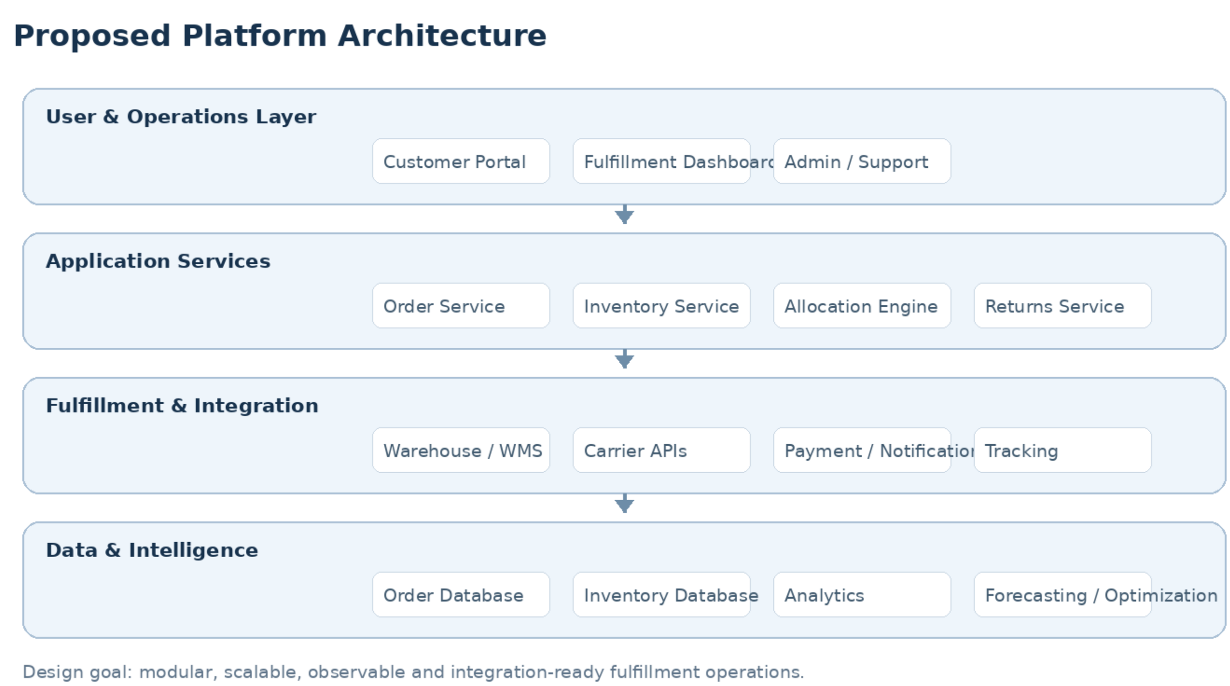
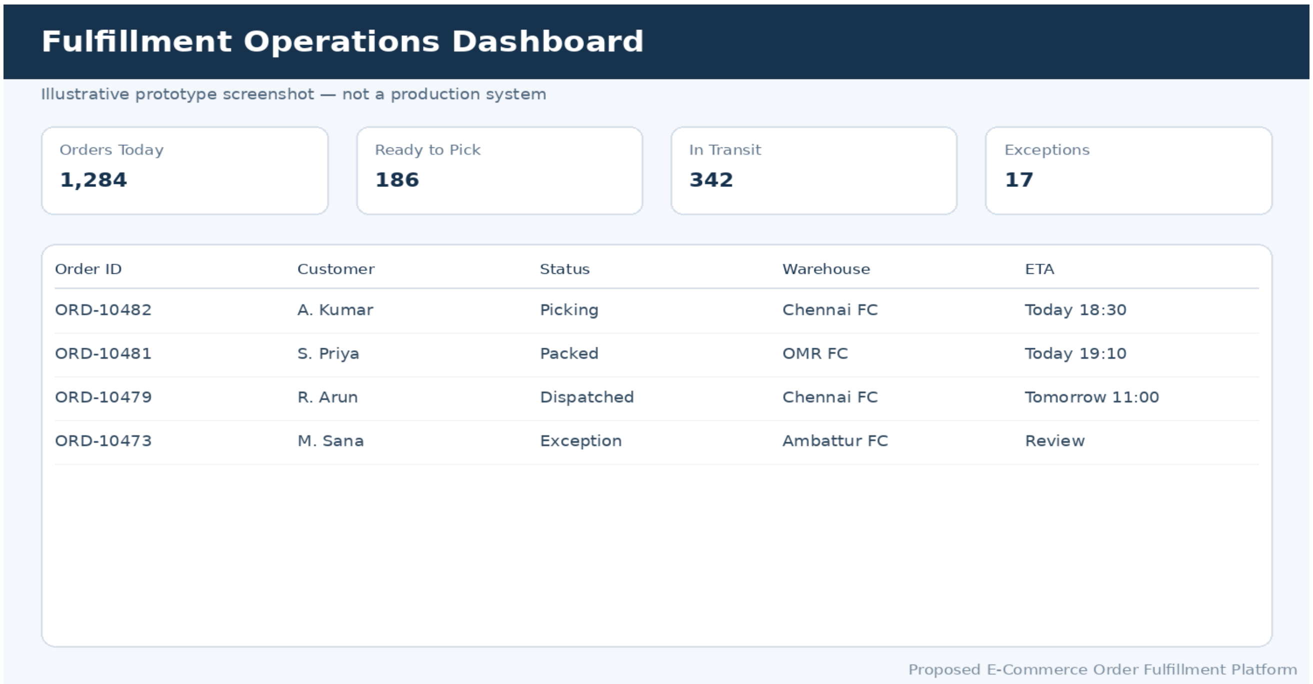


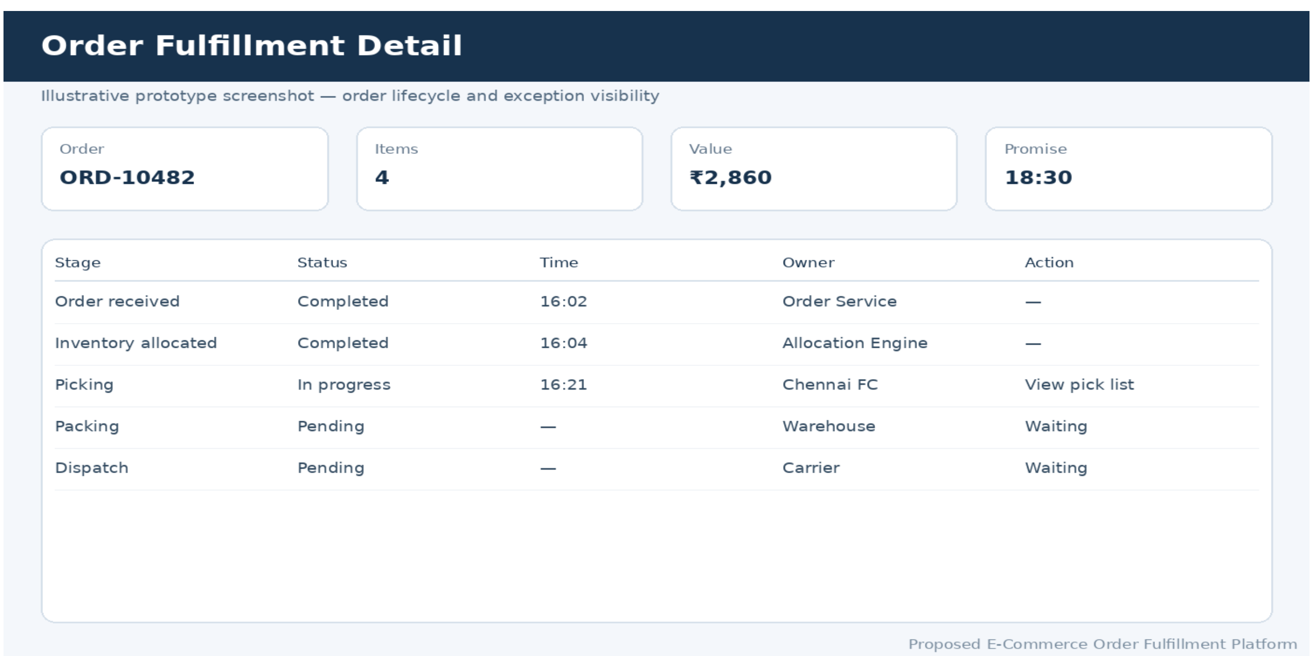
Figure 2. Proposed modular architecture linking customer/operations interfaces, application services, warehouse and carrier integrations, and data/intelligence components.

Prototype Screenshots and Expected Outcomes

Fulfillment Operations Dashboard



Order Fulfillment Detail



Inventory and Allocation Monitor

Inventory & Allocation Monitor

Illustrative prototype screenshot — stock visibility across fulfillment nodes

SKUs	Low Stock	Allocated	Fill Rate
8,420	73	1,964	96.8%

SKU	Product	Chennai FC	OMR FC	Allocation
SKU-1182	Wireless Mouse	124	38	Chennai
SKU-2204	USB-C Hub	19	82	OMR
SKU-3177	Keyboard	7	44	OMR
SKU-4419	Headset	51	11	Chennai

Proposed E-Commerce Order Fulfillment Platform

Shipment and Last-Mile Tracking

Shipment & Last-Mile Tracking

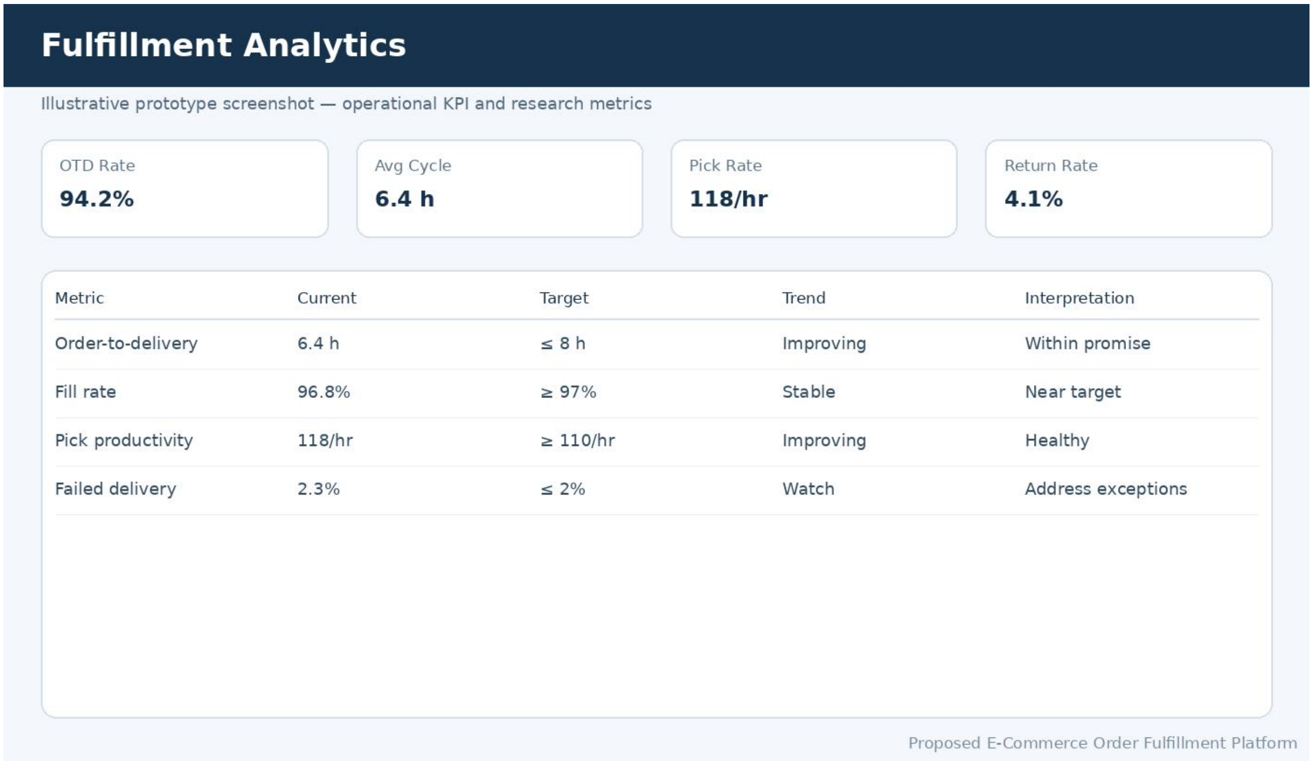
Illustrative prototype screenshot — customer-facing delivery visibility

Dispatched	On Route	Delivered	Failed
342	221	1,021	8

Shipment	Carrier	Current Stage	ETA	Risk
SHP-7821	Partner A	Out for delivery	18:30	Low
SHP-7819	Partner B	Hub scan	20:10	Medium
SHP-7813	Partner A	Out for delivery	19:05	Low
SHP-7802	Partner C	Delayed	Tomorrow	High

Proposed E-Commerce Order Fulfillment Platform

Fulfillment Analytics



Research Gap to Proposed Solution Mapping

Research Gap	Proposed Solution	Expected Benefit
Fragmented stage-wise optimization	Integrated order, inventory, warehouse and delivery services	End-to-end coordination
Static operational rules	Dynamic allocation and batching inputs	Better response to demand changes
Limited exception treatment	Exception service with alerts and reallocation	Faster recovery
Narrow visibility	Unified dashboard and order timeline	Improved operational control

Future Research Directions

- AI-assisted demand forecasting linked directly to inventory placement and fulfillment-center allocation.
- Reinforcement learning or adaptive optimization for dynamic order allocation under changing inventory and carrier capacity.

- Digital-twin simulation of warehouse, middle-mile and last-mile operations before deploying policy changeSocietal and Software-Engineering Impact

Conclusion

The literature demonstrates that e-commerce order fulfillment is a complex, interconnected problem involving inventory, warehouse execution, order batching, allocation, transportation and last-mile delivery. Existing studies provide strong methods for individual decisions, including queuing models, simulation, predictive analytics, vehicle routing, Markov Decision Processes and stochastic optimization. However, the major research opportunity is to combine these insights within an integrated, real-time and exception-aware software platform.

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